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The circulation and water masses of the Antarctic shelf and continental slope between 30 and 80°E

A.J.S. Meijers [a,b,d,*], A. Klocker [a,b,d], N.L. Bindoff [a,b,d], G.D. Williams [c,a], S.J. Marsland [e,a]

[a] Antarctic Climate and Ecosystems Cooperative Research Centre, Private Bag 80, Hobart, TAS 7001, Australia
[b] CSIRO Marine and Atmospheric Research, GPO Box 1538, Hobart, TAS 7001, Australia
[c] Institute of Low Temperature Science, Hokkaido University, Sapporo, Japan
[d] IASOS, University of Tasmania, Private Bag 77, Hobart, TAS 7001, Australia
[e] CSIRO Marine and Atmospheric Research, Private Bag 1, Aspendale, VIC 3195, Australia

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ABSTRACT

The circulation and water masses from the Antarctic continental shelf to 62°S between 30 and 80°E are described using hydrographic data collected on seven hydrographic sections during the Baseline Research on Oceanography, Krill and the Environment-West (BROKE-West) experiment. The eastern limb of the Weddell Gyre dominates circulation between 30 and 40°E, and is significantly cooler and fresher than the region to the east. The Antarctic Circumpolar Current (ACC) extends from the north into the survey region east of 40°E, reaching as far south as 65.5°S at 60°E. This results in increasing observed maximum temperature and salinities progressively towards the east, peaking at 80°E due to the intrusion of the southern ACC Front (sACCF) to 63°S. This southward extension is steered by the southern end of Kerguelen Plateau, causing a horizontal shear of over 0.15 m s^{-1} between the eastward ACC and westward-flowing Antarctic Slope Current (ASC). The ASC is observed at all six meridional sections immediately north of the shelf break. It is strongly barotropic and transports a total of 15.8 ± 7.4 Sv westwards, while the bottom referenced baroclinic component only contributes 1.3 ± 0.3 Sv. At each section this current intensifies to a narrow westward 'jet' with absolute velocities up to 0.3 m s^{-1} over the steepest shelf slope gradients. At 70°E a 'V' shape is observed in the ASF. This, and the nearby presence of denser shelf water and ice-shelf water, is characteristic of Antarctic Bottom Water (AABW) formation, but no new AABW is found on this section. Instead, significantly warmer, saltier and less oxygenated AABW to the east and newly formed AABW high on the continental slope immediately to the west suggest a formation region just west of 70°E. This newly formed AABW progressively becomes warmer and saltier west of 60°E and is observed extending offshore and moving westward below eastward flowing water masses. ACC frontal positions are found to be 1–2° farther north in the survey region than suggested by historical climatology.

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1. Introduction

The region between 30 and 80°E is relatively poorly sampled in comparison with other regions around Antarctica. Complete suites of physical, biogeochemical, biological and ecological measurements, as are described in this volume from the BROKE-West experiment, are especially rare. Yet this region is the site of an important confluence in the polar circulation between the region west of Weddell-Enderby Land, Kerguelen Plateau to the north and the Australian-Antarctic Basin to the east. Additionally the Prydz Bay area has been widely suggested as a region for bottom water formation (Jacobs and Georgi, 1977; Orsi et al., 1999; Yabuki et al., 2006), so merits more thorough investigation.

Geographically the survey region is contained inside the Weddell-Enderby basin, with the Kerguelen Plateau immediately to the north-east and the shallowest point of the Princess Elizabeth Trough (PET) to the east. The survey area can broadly be divided into two regimes; the area west of around 50°E is dominated by the eastward extension of the Weddell Gyre (Gordon, 1998; Park et al., 2001), whilst west of this the Antarctic Circumpolar Current (ACC) and its southern fronts, i.e. the southern ACC front (sACCF) and Southern Boundary (SB) (Orsi et al., 1995), intrude into the domain from the north, are forced southward by the Kerguelen Plateau, and flow eastward through the PET. These regimes are geographically separated by the northward protrusion of Enderby Land and the Cosmonaut Sea. The Amery Ice shelf is also an important feature, feeding shelf and ice-shelf water into the survey region.

* Corresponding author at: CSIRO Marine and Atmospheric Research, Castray Esplanade, Hobart, TAS 7000, Australia.
E-mail address: andrew.meijers@csiro.au (A.J.S. Meijers).

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The water masses south of the eastward flowing ACC are largely advected from the Australian Antarctic Basin east of the PET (Bindoff et al., 2000; Mantisi et al., 1991) by the westward-flowing Antarctic Slope Current (ASC). This current flows unbroken across the survey region and forms the southern limb of the Weddell Gyre (Park et al., 2001). Results from the original BROKE experiment between 80 and 150°E (Nicol et al., 2000) show that the ASC plays an important role in the distributions of krill, cetaceans and chlorophyll in addition to the along-slope water masses (Bindoff et al., 2000). Earlier work along the ASC shows a significant current core just offshore of the shelf break (Wong et al., 1998). However, these earlier observations are hampered by being only measurements of the density field, and so this strongly barotropic current core is poorly determined (Park et al., 2001). Through the use of both ship based acoustic Doppler current profiler (ADCP) velocities and CTD derived density fields, Bindoff et al. (2000) showed that the slope current in the region 80−150°E typically had a westward barotropic transport between 10–40 Sverdrups ($1\,\mathrm{Sv} = 10^6\,\mathrm{m^3\,s^{-1}}$), and a baroclinic transport component of 0–6 Sv westwards (using a surface reference level). Similar magnitudes in the total slope current for the PET immediately east of the BROKE-west region were obtained by Heywood et al. (1999). In both of these studies the barotropic component of the ASC is very strong relative to the baroclinic component, meaning that direct velocity measurements are required to determine the full velocity field (i.e. both barotropic and baroclinic components). The same situation is observed in this study, and we resolve for the first time both the baroclinic and barotropic components of the ASF currents for the coastal region 30−80°E.

Another open question in the regional oceanography is the role of the Enderby Land and Prydz Bay regions in the production of Antarctic Bottom Water (AABW) (denser than $28.27\,\mathrm{kg\,m^{-3}}$, Whitworth et al., 1998). Earlier CTD data attributing bottom water formation from Prydz Bay (Wong et al., 1998) were inconclusive because of the poor quality of the salinity measurements (Bindoff et al., 2003). New mooring data from beneath the main polynya in Prydz Bay show that the salinity of high-salinity shelf water (HSSW) water in winter is not as high as that observed in the Mertz Polynya, and flows beneath the Amery Ice Shelf rather than directly towards Prydz Channel. Together these two observations suggest that Prydz Bay is unlikely to be a source of bottom water. However, the region offshore from Cape Darnley immediately to the west does show high oxygen concentrations and bottom-intensified flows (Thurnherr, pers. comm. 2004) suggestive of local bottom-water formation. A section at 61°E (Baines and Condie, 1998) shows active AABW formation and Yabuki et al. (2006) observe conditions suitable for AABW formation at the western edge of Prydz Bay.

The exchange of AABW between the Weddell-Enderby Basin and the Australian-Antarctic Basin through the PET also remains poorly understood. Earlier work (Rintoul, 1998; Bindoff et al., 2000; Williams and Bindoff, 2003; Marsland et al., 2004; Williams et al., 2008a, b) has shown that the Adélie Land Bottom Water formed in the Australian-Antarctic Basin contributes 12–15% of global Antarctic Bottom Water. Evidence from water-mass properties and CFCs suggests a significant fraction of this water flows westward through the PET into the Weddell-Enderby Basin (Mantisi et al., 1991). North of the westward-flowing Antarctic Slope Current (ASC), the eastward-flowing ACC carries water from the Weddell-Enderby Land basin into the Australian Antarctic basin, and mixes on the eastern side of the Kerguelen plateau with AABW derived from Adélie Land (Heywood et al., 1999).

This paper covers the physical oceanography data collected on the BROKE-West voyage, the fronts and main water masses of the region, the meridional structure of water mass properties, the large scale circulation and transports within the region, and evidence for Antarctic Bottom Water formation.

2. Data and methods

The measurement program was carried out from January 10 to February 28, 2006 as part of a multi-disciplinary survey of the large scale circulation and biology of the East Antarctic coast between 30 and 80°E. This program consisted of acoustic and krill surveys, megafauna surveys, minicosm experiments, and detailed biological sampling of the mixed layer. This experiment complements the analogous BROKE experiment carried out in 1996 along the East Antarctic coastline from 80 to 150°E (Bindoff et al., 2000; Nicol et al., 2000).

The oceanographic component of the survey consisted of 11 north–south sections, joined by a single east–west section along approximately 62°S (Fig. 1). All of the north–south CTD sections cross the shelf break onto the continental shelf and extend northward to 62°S. In this sector the shelf break varies in depth from 200 m to almost 500 m, and can be observed in Fig. 1 immediately inshore of the 500 m isobath. CTD sections were carried out on Sections 1, 3, 5, 7, 9, 11 and 12. There were 120 CTDs in total, with around 13 CTDs on each meridional leg. North of the 2500 m isobath on each leg the stations were spaced at approximately 75-km intervals, while stations south of this were occupied at every 500 m isobath in order to resolve the continental slope and shelf break. Additionally, three CTDs were conducted in the Prydz channel immediately to the east of Cape Darnley. All CTD stations except station 10 were occupied to full depth.

Oceanographic measurements were taken to WOCE accuracy (Saunders, 1991) using a SeaBird SBE9plus CTD (serial 704), with dual temperature and conductivity sensors and a single SBE43 dissolved-oxygen sensor. CTD salinity has an accuracy of 0.002 (PSS78), temperature of 0.001 °C and oxygen concentration is accurate to within 1%. A full technical report is available in Rosenberg (2006). Ship-mounted Acoustic Doppler Current Profiler (ADCP) data were collected as underway data throughout the voyage. However, contamination by ship acceleration and acoustic ringing due to the hull geometry means only data gathered when the ship was traveling at less than $0.35\,\mathrm{m\,s^{-1}}$ were used. The ship 150 kHz ADCP operates with 8-m-bin sizes and pulse lengths, 4-m blanking intervals, pinging once a second. Sixty bins are used, giving a maximum range of 480 m, although the range is typically significantly less than this in practice. Ensemble averaging occurs over 3 minutes intervals, and half-hour intervals during final processing. Headings and position are determined using a ship mounted Ashtech 3D GPS. Post-processed ADCP velocity data have an average error of $0.056 \pm 0.022\,\mathrm{m\,s^{-1}}$.

Two Sontek lowered ADCP (LADCP) operating at 250 kHz were mounted on the CTD frame (one looking upward and one downward) and used on almost all CTD casts to record the current velocity shear over the full depth. This is translated into absolute u and v components using the LADCP mounted compass heading and the vertical integral of the vertical shear, using bottom tracking and the ship ADCP and GPS position as boundary conditions. The LADCP pings at one-second intervals, with a single ping accuracy of $0.056 \pm 0.002\,\mathrm{m\,s^{-1}}$. Bin size and pulse length were both 8 m in the vertical, with a blanking bin size of 4 m. Range varied depending on the *in situ* conditions, but was generally around 100–150 m for each instrument. Post-processing averaged the data into vertical 20-m bins. The incorporation of bottom tracking and the ship ADCP into the profile means that the LADCP velocities in the surface 100–200 dbar and bottom 100 dbar are the most reliable, with root mean square errors of

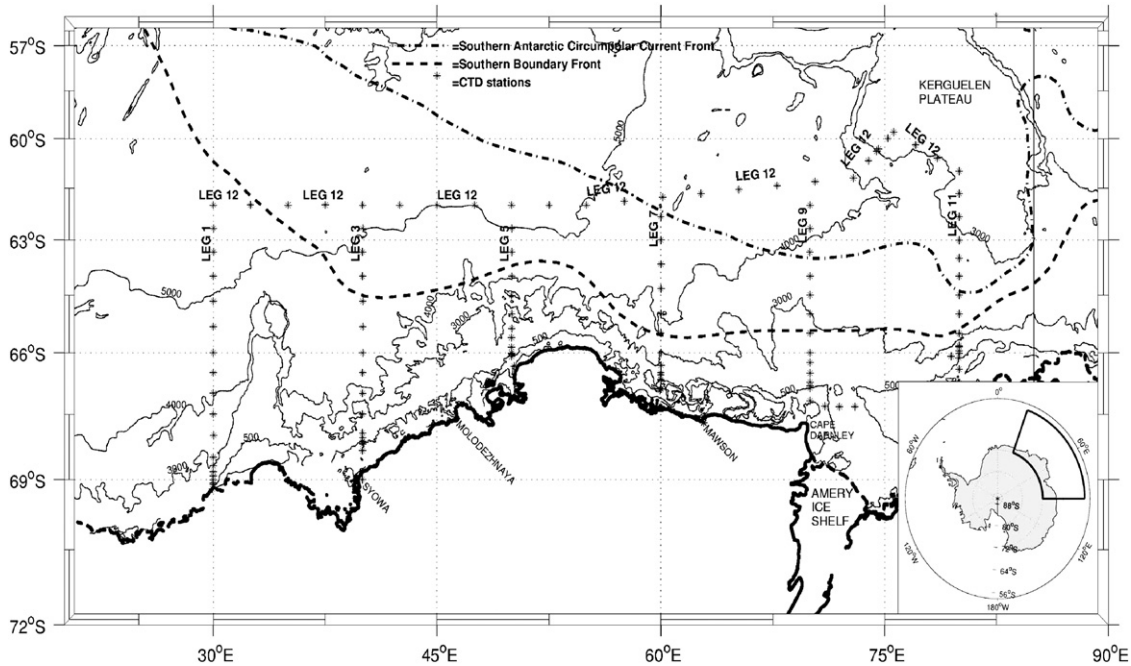

Fig. 1. BROKE-West survey region including bathymetry, CTD stations and major frontal and geographical locations. Frontal positions are from Orsi et al. (1995).

0.028 and 0.030 m s^{-1}, respectively, at these depths. Tides were removed from the LADCP velocities using the CATS02.01 tidal model (Padman et al., 2002), and were found to be small and have a minimal impact on the overall velocity structure of the region.

The absolute velocities estimated in this manner are insufficiently accurate to calculate transports that close the mass transport budget in the survey region. When calculating volume transport into the closed boxes formed by the ship track against the coast, volume imbalances of between 10.6 and 75.6 Sv were encountered using the absolute velocities. These far exceed the divergence that may be expected due to leakage over the continental shelf or due to currents changing during the time taken to complete the transect around each side of the boxes, and are orders of magnitude greater than the baroclinic divergences (0.5–3.2 Sv).

The unrealistically large absolute volume transport divergences are due to the sensitivity of the integration to relatively small LADCP errors, as well as temporal aliasing. The large spacing between stations north of the continental shelf (approximately 75 km) and their depth (> 4000 m) means an error in the LADCP of 0.03 m s^{-1} translates into a 7.2 Sv error in the transport for one station. Added in quadrature around the stations north of the shelf break (approximately 20 for a single box formed by two adjacent meridional transects) this amounts to 32.2 Sv, or 144.0 Sv if the worst case error is assumed at each station.

3. Front and water mass definitions and distributions

There are three main fronts passing zonally through the survey region. From north to south these are the southern Antarctic Circumpolar Current Front (sACCF), the Southern Boundary of the ACC (SB) and the Antarctic Slope Front (ASF). The ASF is the strong subsurface horizontal gradient of temperature and salinity separating the lighter Antarctic Surface Water (AASW) from the denser Modified Circumpolar Deep Water (MCDW), found over the continental slope, and the Circumpolar Deep Water (CDW) further to the north. We define the ASF location using the position of the southern most penetration of the 0 °C isotherm below the winter water, following Ainley and Jacobs (1981) as a southern limit, with the northern limit defined using the Whitworth et al. (1998) definition of a strong horizontal T-S gradient at 200–400 dbar. Orsi et al. (1995) define the SB as the southern extent of Upper Circumpolar Deep Water (UCDW) oxygen-minimum waters, which closely corresponds to the 1.5 °C isotherm in this survey region. The sACCF is defined here as having a temperature maximum of $T_{max} > 1.8$ °C and salinity maximum $S_{max} > 34.73$ psu.

Table 1
Bounding potential temperature, salinity and density values that define the water masses in the BROKE-West region.

	Neutral density (kg m^{-3})	θ (°C)	S (psu)
AASW	$\gamma_n < 28.03$	−1.84 to 2	> 34
CDW	$28.03 < \gamma_n < 28.27$	> 1.5	> 34.5
MCDW	$28.03 < \gamma_n < 28.27$	< 1.5	< 34.7
SW	$\gamma_n > 28.27$	< −1.7	> 34.47
ISW	$\gamma_n > 28.03$	< −1.9	34.3–34.5
AABW	$\gamma_n > 28.27$	> −1.7	> 34.6

AASW is Antarctic Surface Water, CDW is Circumpolar Deep Water, MCDW is modified Circumpolar Deep Water, SW is Shelf Water, ISW is Ice Shelf Water, and AABW is Antarctic Bottom Water.

The water masses observed in this experiment are defined in Table 1. The most important of these are AASW, CDW, MCDW and Antarctic Bottom Water (AABW). The definitions of these water masses follows Whitworth et al. (1998). Throughout the text we use the neutral density variable γ_n (Jackett and McDougall, 1997) for density and whenever we refer to an isopycnal or density it refers to the neutral density.

The water masses defined in Table 1 can be observed in Fig. 2 (All Legs). AABW ($\gamma_n > 28.27$ kg m^{-3}) is found in all of the hydrographic data north of the continental slope, and has a considerably 'thicker' profile in T-S space on Leg 7 than in any other, indicating mixing with surrounding water masses. In contrast the AABW on the eastern Legs 9 and 11 have a very narrow range of T-S values, and are considerably more saline and warmer than the AABW found at Leg 7 and farther west.

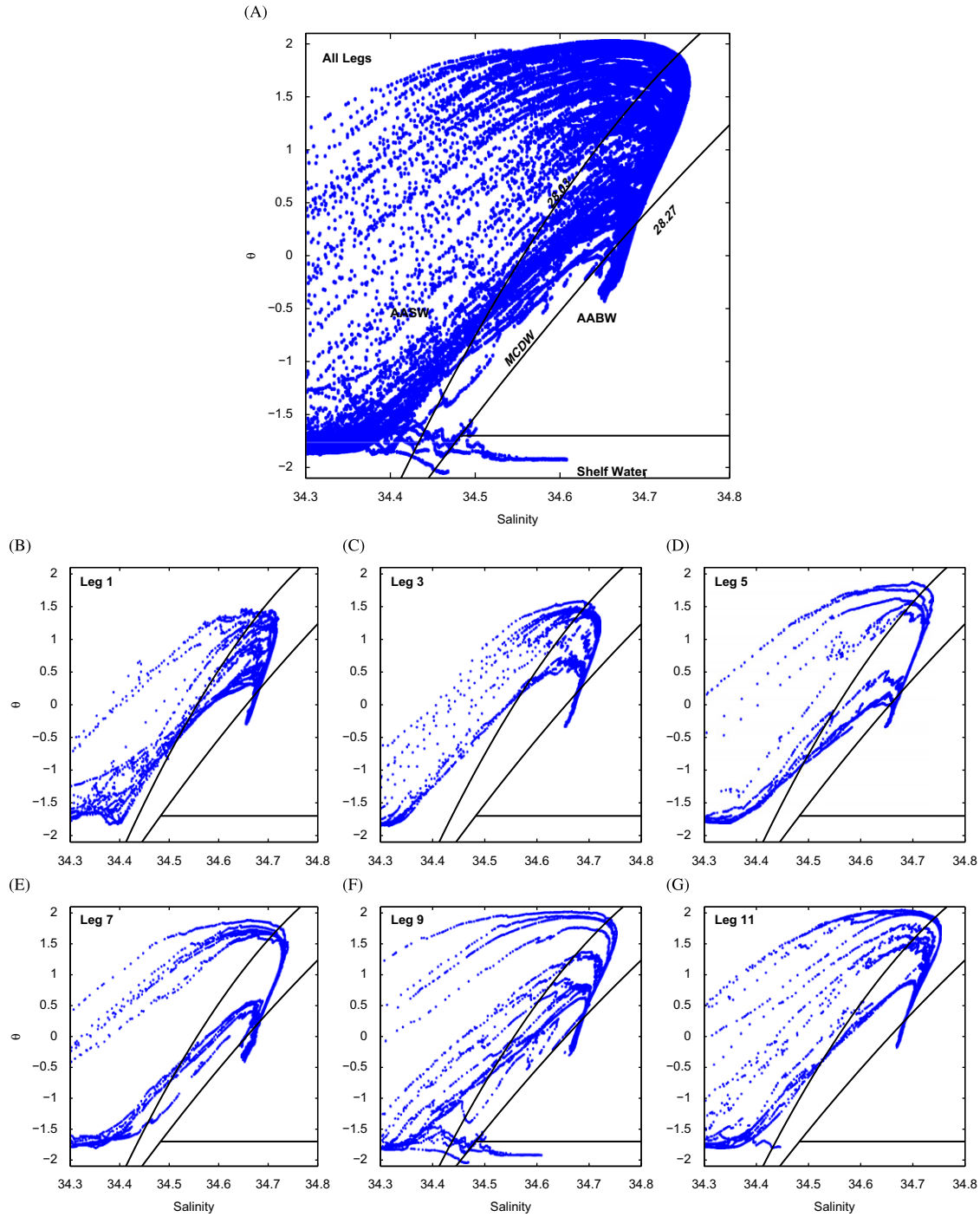

Fig. 2. Temperature salinity plots for all legs combined (all legs) and each individual leg (Legs 1, 3, 5, 7, 9 and 11). The two continuous lines are the 28.03 and 28.27 kg m^{-3} density surfaces defined by Whitworth et al. (1998) that separate the circumpolar watermasses Antarctic Surface Water (AASW), modified Circumpolar Deep Water (MCDW) and Antarctic Bottom Water (AABW) (see Table 1). The horizontal line separates AABW from Shelf Water (SW).

MCDW and CDW is bounded above and below by the density surfaces 28.03 and 28.27 kg m^{-3}, with CDW being warmer and saltier than MCDW. There is a distinct spatial structure to the MCDW and CDW evident in Fig. 2. Each leg has two or more distinct limbs to the T-S distribution, representing the meridional separation of watermasses by fronts. This is most obvious in Legs 5 and 7 where water masses at the same density, horizontally separated by no more than 75 km, may have temperature and salinity differences of over 1 °C and 0.1 psu, respectively. There is also obvious spatial structure between the sections. Legs 1 and 3 have only two distinct limbs to their T-S distributions, separated by the ASF. The cooler limb of these two sections is significantly warmer ($T_{max} > 0.6$ °C) than the equivalent limb of Legs 5 and 7 ($T_{max} < 0.5$ °C), whilst the warmer (offshore) limb has a $T_{max} < 1.4$ °C and $S_{max} < 34.71$, significantly cooler and fresher than any of the other four sections, agreeing with the properties of warm core Weddell Front eddies observed in the region by Gouretski and Danilov (1993) ($0.8 < T_{max} < 1.4$ °C) and suggesting a Weddell Gyre origin for these waters. Legs 5 and 7 have a more spread out three limbed structure in MCDW T-S space than Legs 1

and 3. The separation between the cooler two limbs represents the ASF, while the warmer limb is separated from the other two by the SB, and contains CDW from the ACC. The T_{max} and S_{max} increases across these two legs to greater than 1.85 °C and 34.74 psu, respectively, putting them outside the range defining the Weddell Front ($S_{max} < 34.72$ psu, $T_{max} < 1.5$ °C, Park et al., 2001; Schröder and Fahrbach, 1999). The T_{max} and S_{max} continue to increase further east and a third front, the sACCF appears at Legs 9 and 11. The greatest T_{max} and S_{max} are reached on Leg 11, with $T_{max} > 2$ °C and $S_{max} > 34.75$ psu, agreeing well with the observations of Bindoff et al. (2000) at the same longitude.

Only two water masses are localised and not found across all sections. Ice Shelf Water (ISW), defined as cooler than the surface freezing point of approximately −1.9 °C (Foldvik et al., 2004), is formed through the interaction at depth by Shelf Water (SW) with ice, probably the base of the Amery Ice Shelf in this case (Wong et al., 1998). ISW is observed on the western-most profile of the three CTD casts immediately east of Cape Darnley at the southern end of Leg 9 (Figs. 1 and 2, Leg 9). Denser SW is observed on the other two casts, but at no other location in the survey region. The most saline SW has a salinity of 35.606 psu, and originates from either winter sea-ice formation or possibly onshelf mixing with MCDW and subsequent winter cooling (Wong et al., 1998).

4. Meridional section structure

4.1. Vertical section profiles

We describe three sections (3, 7 and 9) to characterise the meridional structure of the area, water masses near the shelf break and other features of interest. In Section 3 (40°E) two sharp horizontal gradients of temperature, salinity, density and oxygen in the upper 500 m occur near 66.5°S and 68°S (Fig. 3). We label the deeper and southern of these two gradients the northern limit of the ASF. The AASW over the shelf extends all the way to the bottom, and there is no dense shelf water observed on this transect. It is worth noting here that the station spacing north of the shelf break is fairly coarse (75 km) and the frontal features may be narrower than shown in the figures.

The sloping surface isopycnals associated with the ASF drive the surface intensified westward ASC, which has its core at 68°S, forming a jet moving at up to 0.15 m s^{-1} (Fig. 3D). This ASC jet is seen in all sections, and occurs over the steepest gradient of the shelf break. Unlike the ASC observations by Heywood et al. (1998), only a single core is observed at all sections, rather than two. However, the jet observed here corresponds well to the fastest jet observed in their study over the shelf break. This topographic control has also been observed by Muench and Gordon (1995) and Heywood et al. (2004) who observe an ASC jet in the

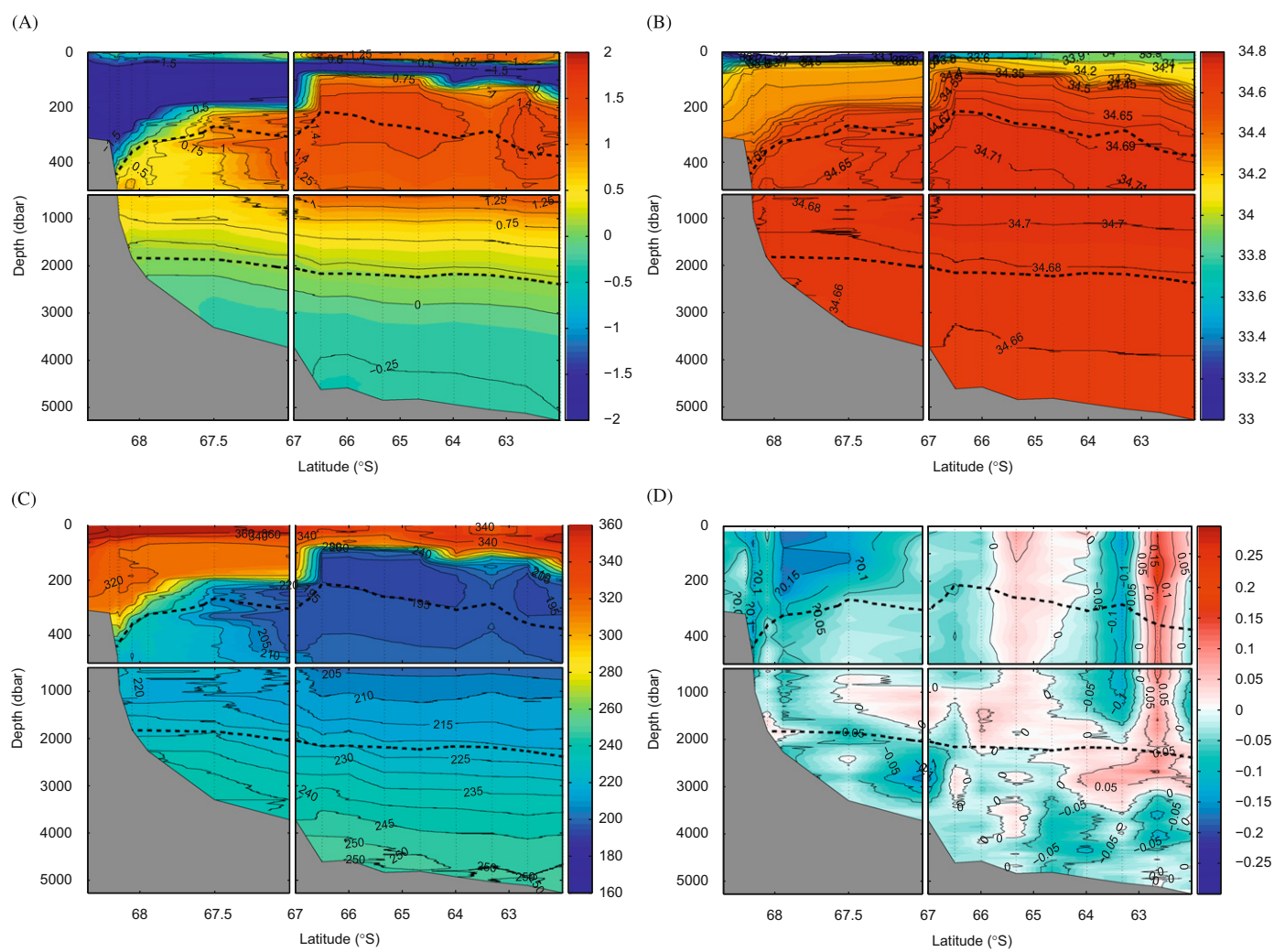

Fig. 3. Leg 3, (A) potential temperature (°C), (B) salinity (psu), (C) oxygen (μmol l^{-1}), (D) LADCP zonal velocities (m s^{-1}), negative is westward. Bold dashed lines represent the 28.03 kg m^{-3} (upper) and the 28.27 kg m^{-3} (lower) neutral surfaces. Scale changes are indicated by the breaks in the axis.

500–1000 dbar range. Heywood et al. (2004) notes that the ASC and jet are distinctly different from the Antarctic Coastal Current (ACoC) found further south over the continental shelf, although in regions where the continental shelf is narrow the ACoC and the topographically controlled ASC are sometimes hard to differentiate (Heywood et al., 1998). As such the ACoC does not appear obviously in the BROKE-West sections as they generally do not extend sufficiently far onto the continental shelf.

At Section 3 the ASC extends north to around 66.5°S where the surface velocities change direction to the east. This change in transport direction is not uniform through the water column, and the net transport of water remains to the west as far north as 63°S. This westward transport north of the ASC is nearly coincident with the step like property gradients in the upper 500 dbar at 67°S and 64°S between which there is locally increased temperature and salinity. This region of Leg 3 shares T-S properties with the northern end of Leg 1 (not shown), which also exhibits similar step like structures in the surface 200 dbar.

The coherent features shared by Legs 1 and 3 lead us to suggest that the westward flow on these legs is the return flow of the eastern edge of the Weddell Gyre, which apparently reaches to between 40 and 50°E. This extension of the Weddell Gyre is also reflected by the T-S properties of these sections that have notably cooler and fresher temperature and salinity maximums (Fig. 2) in the MCDW than in the sections further east. In addition the low-oxygen ($<200\,\mu mol\,l^{-1}$) core of the MCDW penetrates further south in this section than at either Leg 1 or 5, indicating southward advection of older Weddell Gyre waters at Leg 3. This supports the observation based on climatological hydrography by Park et al. (2001) that the Weddell Gyre's eastern limb should extend to at least 53°E before merging with the return flow of the ASC.

Using the Whitworth et al. (1998) definition of AABW (see Table 1), the 28.27 $kg\,m^{-3}$ density surface at around 2000 dbar indicates the presence of a significant volume of AABW at Leg 3 below this isopycnal. The flatness of this density surface and its high angle of intersection with the continental slope indicate that there is no active AABW formation at this section during observations (Bindoff et al., 2000). This AABW has a significant westward velocity, extending well to the north of the continental slope and moving westwards below surface eastward transport in the north of the section. The most oxygenated, coldest and freshest AABW in this section is found immediately north of the base of the continental slope at 66°S and has no apparent connection with the slope region at this section.

In Section 7 (Fig. 4) at 60°E the AABW signature is seen much higher on the continental slope (1500 dbar) and the 28.27 $kg\,m^{-3}$ density surface is raised by the presence of AABW over the continental slope. However, this contour denoting the AABW boundary still intersects the bathymetry at a relatively high angle and temperature and salinity contours are not continuous from the shelf break down the continental slope. This indicates that no

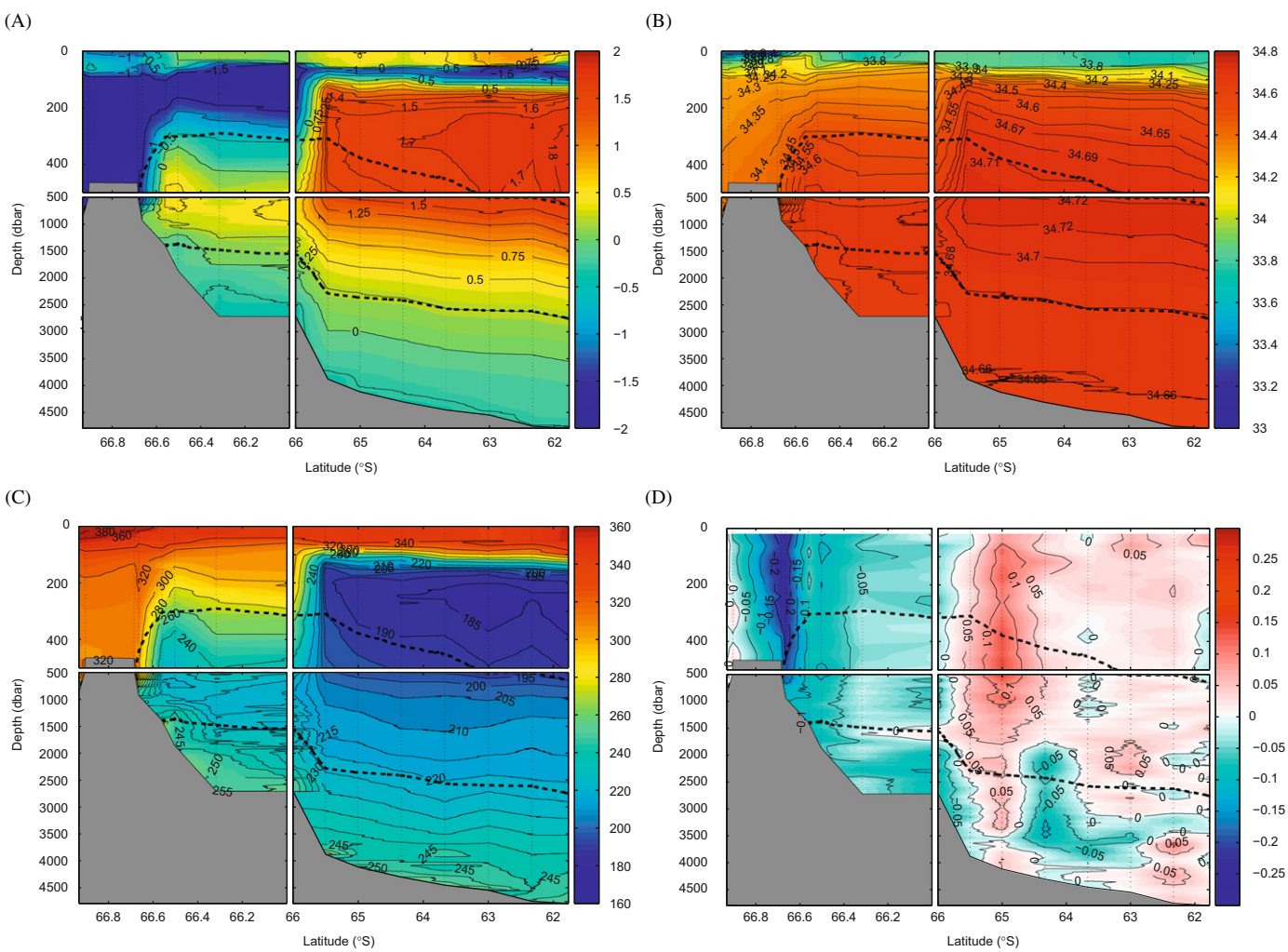

Fig. 4. Leg 7, (A) potential temperature (°C), (B) salinity (psu), (C) oxygen ($\mu mol\,l^{-1}$), (D) LADCP zonal velocities ($m\,s^{-1}$), negative is westward. Bold dashed lines represent the 28.03 $kg\,m^{-3}$ (upper) and the 28.27 $kg\,m^{-3}$ (lower) neutral surfaces. Scale changes are indicated by the breaks in the axis.

AABW is being formed at this section and time. At the shelf break, south of the AABW, the 28.03 kg m^{-3} isopycnal plunges to the bottom at over 500 dbar. This coincides with the southern edge of the ASF and ASC, leading to a strong westward current of over 0.2 m s^{-1}.

The most obvious features of this section are the very sharp ASF at 66.6°S and the SB at 65.5°S. There are strong horizontal gradients in temperature, salinity and oxygen at each of these latitudes, and this section represents the greatest southward extent of CDW in the survey region. Between these two features there is a small upwelling area where warm and salty MCDW intrudes upwards to around 320 dbar. Geostrophically, this intrusion is dominated by the shoaled AABW and the downward to the north tilt of the 28.27 kg m^{-3} density surface. This results in a horizontal velocity difference of almost 0.4 m s^{-1} between the westwards ASC jet (> 0.2 m s^{-1} west) and eastward flow north of the SB (> 0.1 m s^{-1} east) that occurs over a distance of 200 km. As in the previous section there is net westward transport over the entire ASC (Fig. 4D) despite weak gradients in the isopycnals between the southern edge of the ASF and the SB that imply eastward baroclinic transport. The strong eastward ACC north of the SB is very clear in the LADCP data, as is the westward-flowing AABW undercutting the ACC at depth.

Sections 9 (Fig. 5) and 11 (not shown) are distinctly different from the four western CTD legs. At both of the sections the 28.27 kg m^{-3} density surface gradually deepens towards the north, and the 28.03 kg m^{-3} density surface is deep at the shelf break, shoals to the north, and then rapidly deepens again to below 500 dbar at 64°S. The rapid deepening to the north is associated with the presence of the SB and sACCF near 63.5°S and 63°S, respectively, indicating the intrusion of the ACC.

There is a distinct 'V' shape in the 28.03 kg m^{-3} density surface at the shelf break centered at the 4th station from the coast. The salinity figure (Fig. 5B) shows that the onshore side of the horizontal 'V' density gradient is characteristic of the presence of intruding MCDW at the southern end of the section (Ou, 2007). This intruding wedge of MCDW onto the continental shelf is not seen on any other leg. The relatively high-density water also extends in a tongue down the continental slope to almost 1000 dbar, but not to AABW depths. This, and the high angle of incidence of the 28.27 kg m^{-3} density surface with the continental slope, indicates that no AABW is being formed in this section during the observation period. However, the intrusion of MCDW onto the shelf here, and the presence SW and ISW immediately east of the southern end of this section (see Section 3) suggests that seasonal AABW formation at or close to this section may be possible, as the presence of MCDW and SW are necessary for AABW formation (Whitworth et al., 1998).

The westward ASC jet at 66.8°S on Leg 9 is bottom intensified, as the westward barotropic transport (see Section 5.2) is opposed by eastward baroclinic flow due to the sloping density surfaces of the MCDW over the shelf (Fig. 5D). Consequently the strength of

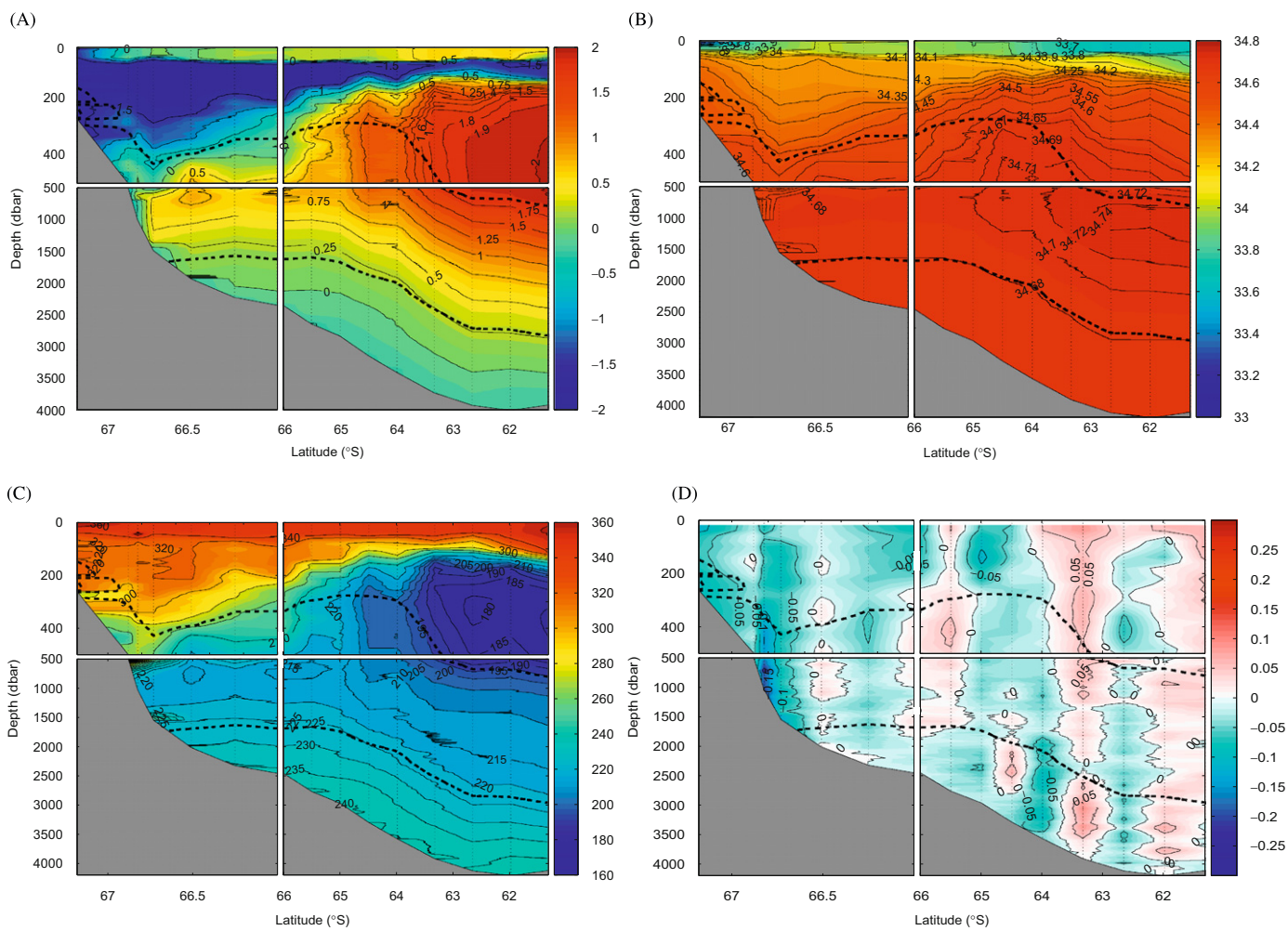

Fig. 5. Leg 9, (A) potential temperature (°C), (B) salinity (psu), (C) oxygen (μmol l^{-1}), (D) LADCP zonal velocities (m s^{-1}), negative is westward. Bold dashed lines represent the 28.03 kg m^{-3} (upper) and the 28.27 kg m^{-3} (lower) neutral surfaces. Scale changes are indicated by the breaks in the axis.

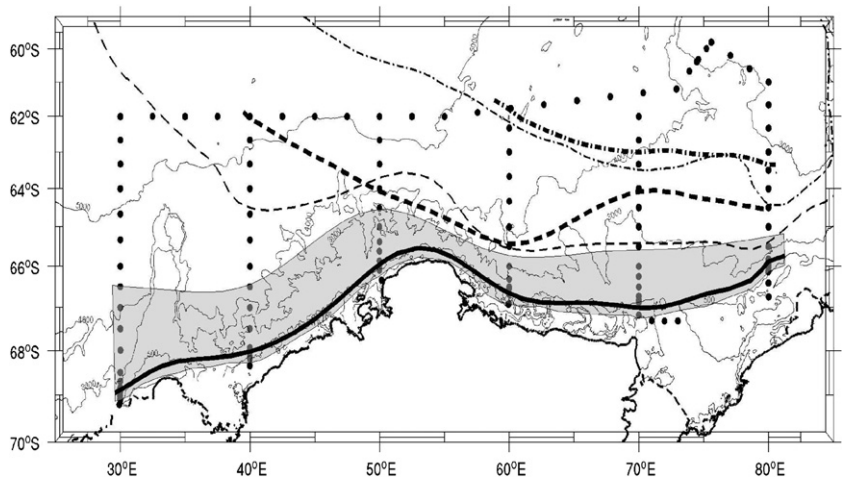

Fig. 6. Frontal features: the dash dotted line is the sACCF, dashed line is the SB, the solid line is the ASC jet and the grey region represents the extent of the ASF. BROKE-West observations are bold, and the Orsi et al. (1995) sACCF and SB positions are the finer lines.

the jet is reduced from over 0.15 m s^{-1} at the bottom to 0.05 m s^{-1} near the surface. Immediately to the north there is a narrow band of eastward transport extending over the full depth range. The cause of this eastward flow is unclear, but may be an eddy spun off the ASC jet. In this section there is a broad region of relatively slow (<0.05 m s^{-1}) westward transport north of the ASF, extending to around 64°S. This region of westward baroclinic transport is significantly broader than in other sections, and is coincident with a greater northward extent of oxygenated waters in the upper 500 dbar than is observed elsewhere in the region. This may represent the western edge of the clockwise circulating Prydz Bay gyre, drawing oxygenated shelf surface waters further offshore to where the isopycnals deepen and low oxygen Upper CDW intrudes north of 64°S. Unlike Sections 3, 5 or 7 there is no signal of westward-flowing AABW beneath the eastward flowing ACC.

4.2. Fronts

The southernmost frontal feature in the survey region is the ASF. Using the Ainley and Jacobs (1981) and Whitworth et al. (1998) definition of the ASF we see that this front broadly follows the shelf break and has its greatest extension away from the coast at Legs 1, 3 and 9 (Fig. 6).

The ASC jet immediately north of the shelf break is a significant feature, particularly for biology, due to its very strong, coherent structure in the velocity field. In each section this extends from the surface to the bottom over the maximum gradient in the shelf break. In most sections this corresponds to depths of approximately 500 dbar, but in Sections 9 and 11 the core of the jet is found at 1000 dbar, agreeing well with observations by Muench and Gordon (1995). There is a very strong barotropic component to the jet, resulting in the highest velocities observed in the survey area, of up to 0.3 m s^{-1}. This strong barotropic component is observed in other studies in the surrounding regions (Heywood et al., 1998, 1999; Bindoff et al., 2000), demonstrating that it is a coherent feature from at least 15°W to 150°E.

The SB is an important feature biologically and represents a significant boundary in primary production (Hiscock et al., 2003). Using the Orsi et al. (1995) definition we see that the SB lies close to the ASF in Sections 5,7 and 11, and rapidly diverges to the north relative to the ASF in the west of the survey region. At Section 9 there is a considerable distance between the ASF and SB, probably due to the northward circulation of the western edge of the Prydz Bay gyre. The SB appears substantially further north in BROKE-West than is found by Orsi et al. (1995), and only on Legs 5 and 7 is there good agreement between the two frontal positions (Fig. 6).

The sACCF is only observed in the eastern part of the survey region. This water intrudes from the north at 60°E, at the northern extreme of Leg 7 and crosses Legs 9 and 11 just south of 63°S. Again this front appears further to the north than in climatological estimates of the frontal position (Orsi et al., 1995). In Legs 7 and 9 the difference is small, but on Leg 11 the sACCF is more than 1° of latitude further north.

The significant coherent northward displacement of the SB and sACCF over much of the survey region from those estimates made by Orsi et al. (1995) based on historical data is perhaps unsurprising. Sokolov and Rintoul (2002) showed that ACC fronts, when not constrained by topography, can meander north and south over several degrees of latitude whilst maintaining their structure. Furthermore Sokolov and Rintoul (2007) showed using altimetric analysis that the sACCF position is highly variable and it may range over a band of latitudes almost 5° wide immediately to the east of the survey region. Therefore the departures of the frontal positions from the smoothed historical climatology observed here may be expected, particularly given the reduced topographic control of the fronts over the Enderby Abyssal Plane, and reflects the natural variability of these features.

5. Large scale circulation and transports

5.1. Surface circulation

To visualise the large-scale surface circulation regimes in the survey region a surface dynamic topography was estimated using both the CTD and LADCP data. A constrained least-squares method that ensures no normal flow at the coast was used to create the surface height field. This topography includes the SSH gradients that are due to both the baroclinic and barotropic flow fields. Details of the creation method can be found in Appendix A.

The surface-height plot (Fig. 7) with an overlying schematic of the circulation based on the height field shows that the surface flow can be split into a western and an eastern part. In the west of the survey region the eastern limb of the Weddell Gyre is seen. The Weddell Gyre extends past Legs 1 and 3 and close to the continent it joins the ASC, which acts as a 'western boundary' for

this gyre returning its flow westwards. At approximately 70°E (Leg 9) there is another clockwise gyre, the Prydz Bay gyre. The ACC and ASC define this gyre's northern and southern boundaries, respectively. The intrusion of the ACC is evident through the high surface height in the northern sections of Legs 5, 7, 9 and 11. Sections 5 and 7 have only small surface height gradients and sit between the two distinct gyres. This separation is probably linked to the protrusion of Enderby Land that splits the survey region into the Weddell-Enderby basin to the west and Prydz Bay to the east. The westward flowing ASC associated with the ASF can be seen across the full zonal extent of the region as a gradient in surface height close to the coast.

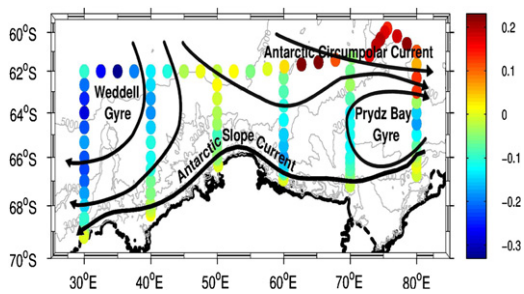

Fig. 7. Sea surface height field shown as coloured circles. Heights are given relative to the stations closest to the coastline, which have a height of 0 m. The overlaid schematic large scale circulation indicates the major regional flow features, following approximate streamlines in the height field.

5.2. LADCP derived transports

The LADCP velocities averaged over the top 200 dbar and bottom 100 dbar, respectively (Fig. 8) reveal key features of the large scale circulation. Immediately obvious is the strong westward transports at the southern end of each leg associated with the ASC, in both the surface and bottom layers, indicating a strong barotropic current. The surface and bottom vectors follow the bathymetry with magnitudes greater than 0.05 m s^{-1} and up to 0.3 m s^{-1} in the ASC jet immediately north of the shelf break (approx. 500 m isobath). In this jet the flow direction is relatively uniform in both bottom and surface flows and the bottom intensification observed further to the east in the Australian Antarctic Basin (Bindoff et al., 2000) is only apparent on Legs 5 and 9. In fact most profiles show slightly weaker westward velocities at the bottom when compared with the surface. At the surface of each leg the westward component of the flow turns eastwards near the northern edge of the ASC or at the center of the Weddell or Prydz Bay gyres in the case of Legs 3 and 9, respectively.

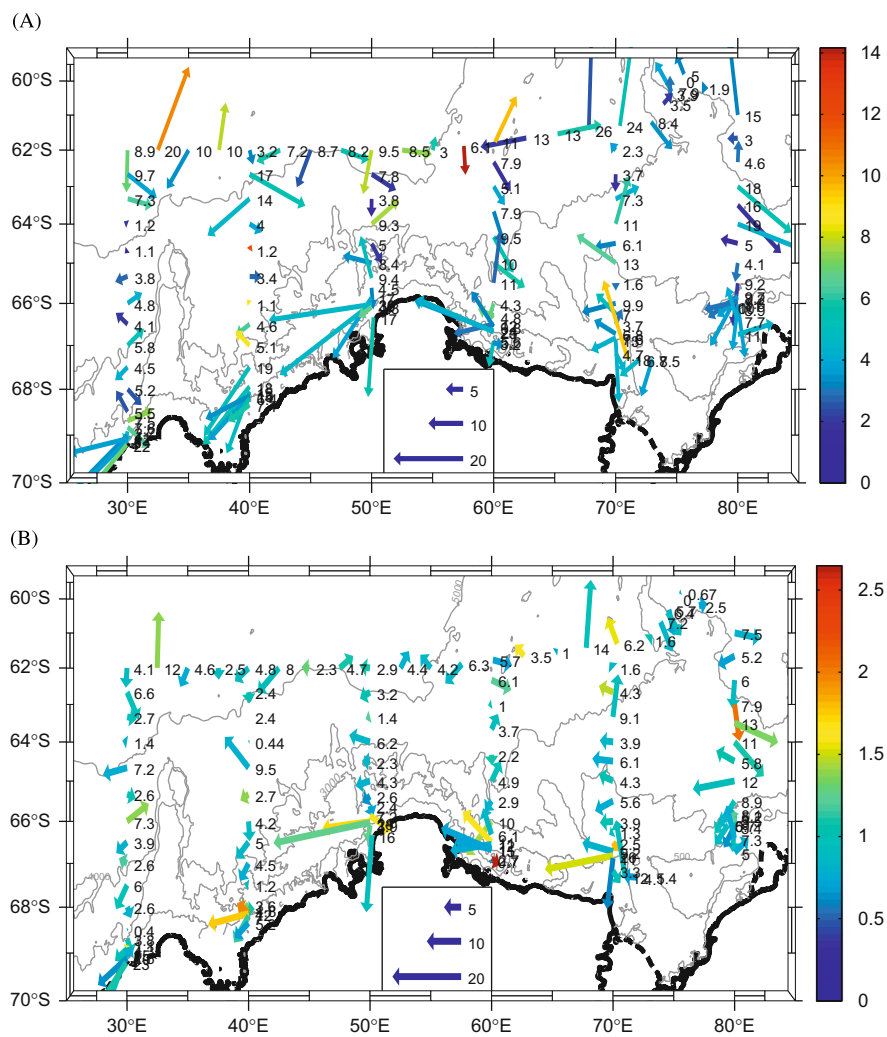

Fig. 8. (A) Surface 200 dbar averaged LADCP velocities and (B) bottom 100 dbar averaged LADCP velocities. Absolute velocity magnitude is labelled for each station, and is also reflected by the length of the vectors. The velocity error is indicated by the colour of the vectors. Units are in cm s^{-1}.

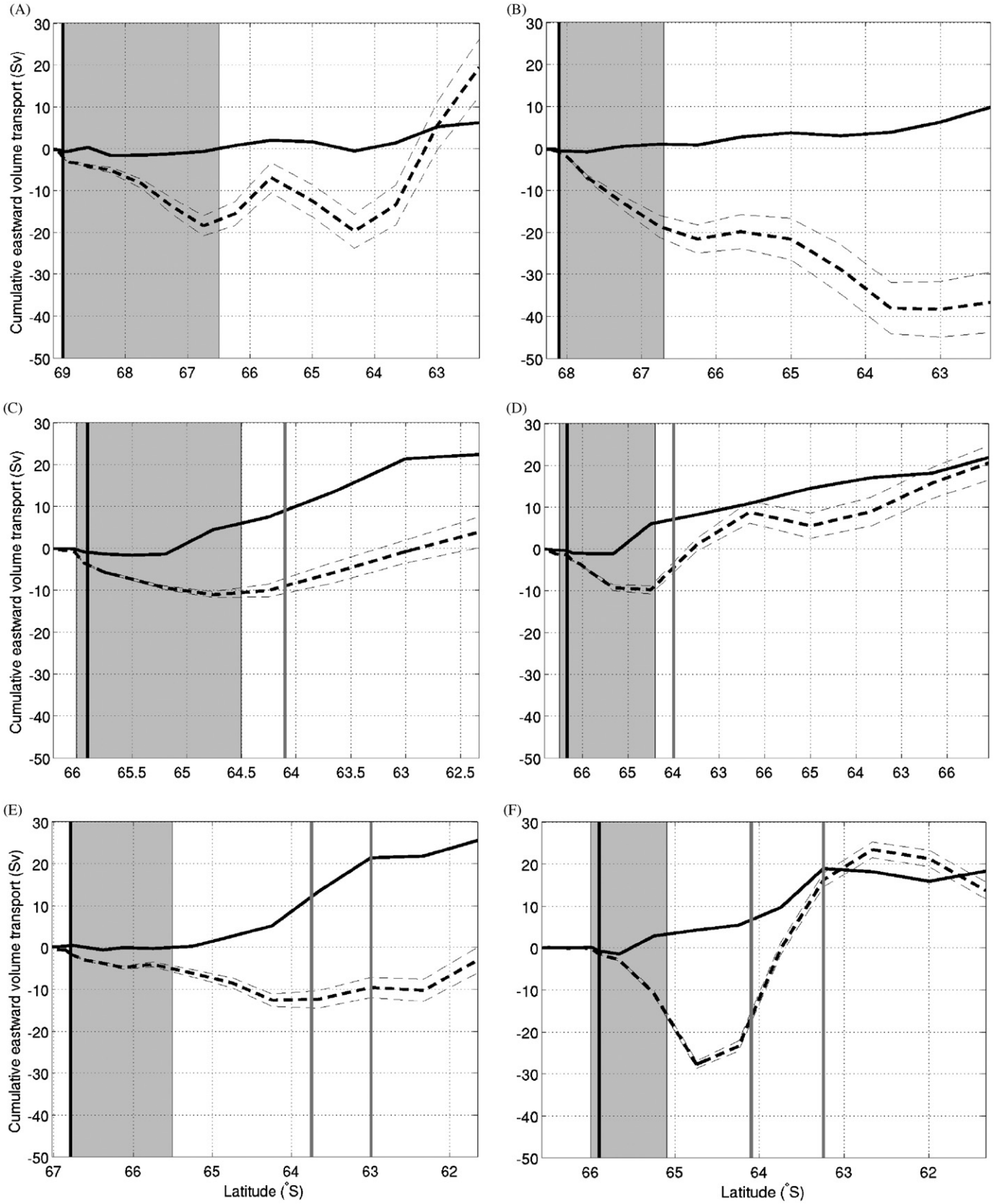

Fig. 9. Cumulative from the south volume transport across each section for (A) Leg 1, (B) Leg 3, (C) Leg 5, (D), Leg 7, (E) Leg 9 and (F) Leg 11. The bold black line gives bottom-referenced baroclinic transport, while the bold black dashed line gives LADCP derived absolute transport. Fine dashed lines indicate 95% error bounds on the absolute transport. The vertical grey region indicates the extent of the ASF and the vertical black line the position of the ASF jet while the next two grey vertical bars indicate the positions (from the south) of the SB and sACCF. Units are Sv.

On all legs except 1 and 3 this westward flow is coincident with the SB of the ACC. Leg 1 has variable LADCP velocities both at the surface and at depth, switching direction at almost every successive station north of the shelf break. There is no obvious indication of inertial oscillations in the ADCP data collected at the individual stations (occupied for over 4 hours), leading us to suggest that localised eddy activity may cause the current switching on Leg 1. This is supported by Schröder and Fahrbach (1999) who observed an intense mesoscale eddy field in the region between 15 and 30°E.

The change in flow direction near the SB at the surface is not seen in the bottom layer. Most sections, particularly on Legs 3 and 5, have more consistently westward bottom velocities than the surface, often as far north as 62°S. The resulting velocity differences between the surface and bottom are frequently greater than $0.1\,\mathrm{m\,s^{-1}}$. These strong bottom transports are consistent with dense AABW sinking and spreading northward and westwards below the eastward flowing ACC. See, for example, the vertical shear in Figs. 3D and 4D around 64.5°S.

Estimates of the bottom-referenced baroclinic and absolute components of the cumulative, depth integrated zonal volume transport across each of the legs are given in Fig. 9. These transports were calculated from the LADCP data and also the CTD density data, as discussed in Section 2, and are shown as cumulative sums from the southern end of each section. Error bars are based on the LADCP error estimate for each cast added in quadrature from the south.

Immediately obvious is the large difference between the bottom-referenced baroclinic transport and the absolute transport, particularly in the case of westward transport in the ASF region. In the ASF (grey shaded region) there is a mean bottom referenced westward baroclinic transport of 1.3 ± 0.3 Sv, while north of the change of current direction from west to east there is on average 18.6 ± 7.7 Sv of eastward baroclinic transport. In contrast the absolute transport has a substantially greater westward contribution in the ASF with a mean of 15.8 ± 7.4 Sv westwards (excluding the anomalous Leg 3) with an approximate error of 3.3 Sv, while the eastward contribution north of this is 26.8 ± 14.0 Sv (error 9.7 Sv). This difference in transports clearly demonstrates that the barotropic contribution dominates the transport of the ASC over the continental slope.

The inclusion of the barotropic component of transport to the ASC extends the region of westward transport significantly farther offshore than in the purely baroclinic case, notably on Legs 3, 5, 7 and 11. Although the baroclinic component of the ASC is strongly surface intensified due to the onshore Ekman transport driving the surface density gradient (Deacon, 1937), below this surface layer and north of the shelf break, the isopycnals slope in the opposite direction (e.g. Fig. 3A) which results in an eastward baroclinic current at depth. This eastward current was also observed by Heywood et al. (1998) immediately to the west. Therefore north of the strong surface baroclinic gradient over the shallow shelf break the deep eastward transport is dominant, and there is a net depth integrated baroclinic eastward transport. When the barotropic component is added, however, the strongly westward barotropic flow in the ASC (up to $0.3\,\mathrm{m^{-1}}$) dominates the relatively weak baroclinic flow ($<0.03\,\mathrm{m^{-1}}$), resulting in the region of net absolute westward transport over the slope extending much farther north than for the purely baroclinic case.

The net eastward transport found north of the SB reflects the intrusion of the ACC into the survey region whilst the large variance of this eastward transport (7.7 Sv baroclinic, 14.0 Sv absolute) is due to the increasingly great southward penetration of the ACC from west to east, with consequently greater eastward transport that strengthens from 15.1 ± 4.0 at 50°E to greater than 40 Sv at 80°E. Other features of the large scale circulation are also apparent in the transport data. The transport of the ASC is greatest at 80°E (27.6 ± 1 Sv westwards), and becomes steadily smaller towards the west, reaching a minimum at 50°E of 9.7 ± 1 Sv westwards, and then increases again to 18.4 ± 2.4 Sv at 30°E. The weakening from 80 to 60°E by 18 Sv reflects the loss of the waters recirculated in the Prydz Bay gyre, while the increase at 30°E indicates the addition of Weddell Gyre waters to the ASC. The exact contribution of the Weddell Gyre to the increased westward transport is difficult to estimate, as at Leg 3 the absolute transport shows westward transport (38.2 ± 6.3 Sv) extending almost to 63°S, while at Leg 1, there is an eastward transport of 18.4 ± 2.4 Sv. This does not appear to conserve mass, as westward transport added to the ASC at 40°E should also appear at 30°E, even allowing for the presence of a strong (11.2 ± 1.7 Sv) warm core eddy centered at 65.5°S on Leg 1 (that appears in both the baroclinic and absolute transport).

A similar conservation problem appears at Leg 9, where there is considerably weaker absolute transport north of the ASF than at either of Legs 7 or 11. Both Legs 3 and 9 are also differentiated from the other four legs north of the ASF in that the absolute transports are in the opposite direction to the baroclinic component. This indicates that the LADCP derived absolute transports are unreliable in the offshore regions north of the slope, as discussed in Section 2. Additionally the AVISO combined mission satellite altimetric product shows sea surface height changes of over 10 cm in less than a 2-week period at the northern edge of the survey region, meaning that there are strong temporal variations of the absolute transport on relatively short time scales in this region. This variability may consequently alias the absolute transport due to the time taken to complete each section and produce the non-conservation of mass observed on Legs 3 and 9 and discussed in Section 2.

6. AABW

Fig. 10 shows the water properties at the bottom of all casts, typically less than 10 m from the sea floor. The temperature and salinity becomes warmer and more saline from west to east, with a distinct jump between Legs 7 and 9 where values change from $T_{min} < 0.3$ °C and $S_{min} < 34.64$ psu to the west to $T_{min} < 0.2$ °C and $S_{min} < 34.66$ psu east of Leg 7. This trend is also apparent in the T-S figures by section (Fig. 2). Additionally there is a north–south gradient, where stations very near the coast tend to be warmer and saltier, become cooler and fresher to the north where AABW (<0 °C) is observed, and then increase in temperature and salinity again near the ACC. These trends are reflected in the oxygen fields where the higher temperature and salinity areas are associated with lower oxygen values. This indicates a greater elapsed time since formation and hence greater mixing with the warmer and more saline overlying MCDW and CDW.

There are, however, several anomalies to these broad trends. Leg 7 shows high concentrations of oxygen, low salinity and low temperature compared to the other legs. Similar properties can be seen at the coastal end of Leg 9. High oxygen concentration values (over $255\,\mu\mathrm{mol\,l^{-1}}$) indicate a newly formed water mass, recently exposed to the atmosphere. Combined with the presence of shelf water and ice-shelf water close to Leg 9 this suggests seasonal AABW production in this region. The LADCP data from Legs 5 (not shown) and 7 (Fig. 4D) show a westward-flowing region of AABW high on the continental slope. This westward flow at the bottom is found progressively deeper (further north) to the west and occupies the abyssal floor in Legs 1 and 3 (Fig. 3D), suggesting AABW is produced east of Leg 7, moves down the slope and is deflected westward due to the Coriolis force (Gill, 1973). This water mass then mixes with the ACC and Weddell Gyre waters

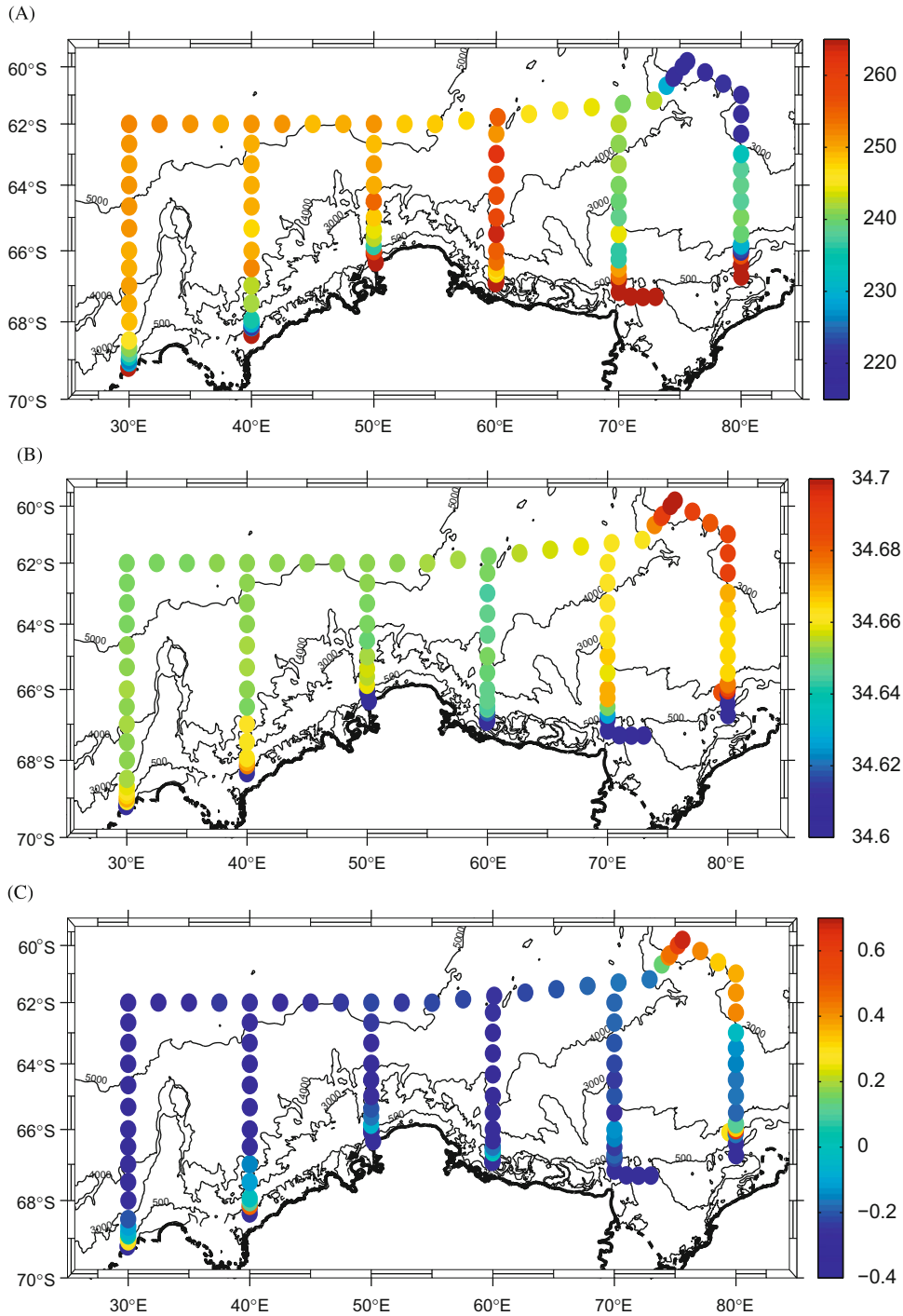

Fig. 10. Water mass properties at the deepest point of each CTD profile (within 10 m of the bottom) for (A) oxygen ($\mu mol\,l^{-1}$), (B) salinity (psu) and (C) temperature (°C).

above it as it moves westward across Legs 1, 3 and 5, eroding the strong characteristics observed on Leg 7.

This evidence for the local formation of AABW is further supported by the presence of significantly lower oxygen, saltier and warmer abyssal AABW in Legs 9 and 11 than in any of the western legs. Even the AABW in Leg 1, which has had the greatest time to mix with the overlying waters still has an oxygen content greater than 250 $\mu mol\,l^{-1}$, in comparison with Legs 9 and 11 which are generally less than 240 $\mu mol\,l^{-1}$. This means there is no strong evidence for the export of newly formed AABW across 80°E, suggesting that the new AABW observed at Leg 7 does not recirculate in the Prydz Bay gyre, and is instead exported to the west. The lack of dense shelf waters and the depth (>1500 dbar) of the AABW plume in Leg 7 suggest that this bottom water appears to originate from the Cape Darnley region, between Legs 7 and 9. Additionally the bottom-water properties close to Cape Darnley suggest AABW production is influenced by the SW and ISW originating from the Amery Ice Shelf, as suggested by Yabuki et al. (2006).

The low oxygen, salty and warm water masses of the ACC intruding from the north on Leg 12 east of 70°E and in the northern part of Leg 11 differ from the position of the ACC seen on

the map of dynamic height (Fig. 7), where the ACC signature appears at around 60°E. This very low oxygen AABW with respect to the rest of the ACC in the survey domain is found north of sACCF and corresponds with westward transport around the south of Kerguelen Plateau on Leg 11. This represents significantly older, warmer and saltier AABW that is advected into the survey region from farther north in the ACC by this Kerguelen counter current.

7. Discussion

The patterns of sea-surface height, water-mass structure and mass transport all suggest that the region may be broadly divided into three main regimes (Fig. 7). Legs 1 and 3 straddle the eastern extension of the clockwise Weddell Gyre, while Legs 9 and 11 are more strongly influenced by the warmer, saltier and lower-oxygen water advected in from the east as part of the ACC, as well as the Prydz Bay gyre that sits between the ASC and ACC. Legs 5 and 7 sit between these two distinct regimes and have newly formed AABW high on their continental slopes.

The distinct difference between Legs 1 and 3 from the more eastern sections is evident throughout the data. The only strong frontal feature in the T-S diagrams of these legs is the ASF, as both the SB and sACCF are found north of these sections. The T-S diagrams for Legs 1 and 3 also have a more limited range in T-S space, with less separation across the ASF than in other sections and they have substantially cooler (0.25 °C) and fresher (0.02 psu) temperature and salinity maximums than any of the more easterly sections.

The cyclonic nature of the eastern limb of the Weddell Gyre means that it turns southward and joins with the westward flowing ASC over Legs 1 and 3. The presence of the eastern edge of the gyre straddling Leg 3 is supported by the LADCP surface velocities (Fig. 8A) which are southeast in the north and southwest further south. The weak zonal flow in the surface layers from 64 to 66°S indicate that Leg 3 may represent the easternmost extent of the Weddell Gyre in the survey region. The strong return flow of the gyre is evident in Leg 3 north of the ASF, increasing the westward transport across the section by almost 30 Sv relative to Legs 5 and 7 (total of 38.2 ± 6.3 Sv). This large value is questionable, especially in light of the known LADCP inaccuracies and the lack of equivalent transport on Leg 1, although this may be due to the presence of transient eddies on this leg. However, it is smaller than some estimates of Weddell Gyre return flow at 0°E of up to 66 Sv (Schröder and Fahrbach, 1999) and provides evidence for the Weddell Gyre extending past 40°E, and that the return flow is dominated by the barotropic contribution, as suggested by Park et al. (2001).

Legs 5 and 7 are distinguished from Legs 1 and 3 by the ACC intruding from the northwest, and from Legs 9 and 11 by evidence of locally formed AABW. These two sections also have considerably 'thicker' AABW T-S profiles (Fig. 2), indicative of mixing caused by recently formed bottom water (see Section 6). Interestingly, whilst the presence of the ACC increases the maximum temperature and salinity of the sections, the ASC limb of the T-S diagram has a type of MCDW that is appreciably cooler and fresher than in Sections 1 and 3 (Fig. 2, cf. Leg 5 with Leg 3). This difference is probably due to the mixing of the ASC water with warmer and saltier Weddell Gyre return flow advected from offshore in Legs 1 and 3.

Legs 9 and 11 are distinct from the rest of the survey region in several ways. Due to the steering effect of Kerguelen Plateau the sACCF is deflected south into the survey domain, and here the CDW is the warmest and most saline in the region. At leg 11, the bottom referenced baroclinic transport is reduced by 3–6 Sv relative to Legs 5, 7 and 9, consistent with some of the flow turning northward at Kerguelen Plateau (Wong et al., 1998). Additionally, the reduced sea surface height at around 65°S on Leg 11 (Fig. 7), the north-eastward surface velocities (Fig. 8) and weaker westward ASC transports of Legs 7 and 9 relative to Leg 11 suggests that there may be a northward cyclonic recirculation of the ASC eastward of around 60−70°E, forming the Prydz Bay gyre. This pattern is quite different from common dynamic height maps, e.g. Wong et al. (1998), which show the Prydz Bay gyre straddling the continental shelf break. This is possibly due to the absence in these maps of the strongly barotropic ASC, as they are based on steric SSH.

On Leg 9 enhanced MCDW mixing is observed in Fig. 5, where a cold, high salinity tongue sinks to below 1500 dbar and in the T-S diagram (Fig. 2) where there is 'thickening' of the southernmost MCDW in T-S space. The intrusion of MCDW onto the shelf at this Leg, and the nearby ISW and SW with a salinity greater than 34.6 psu moving north westward (from LADCP data, not shown) along the western side of the Prydz channel show that the necessary conditions for AABW production (Whitworth et al., 1998) are met in the region. This indicates that the newly formed AABW observed high on the continental slope on Leg 7 probably originated near Cape Darnley as a result of intruding MCDW mixing with Amery Ice Shelf SW and ISW transported into the region along the Prydz Bay channel.

The ASC is shown to be continuous across the survey domain, and its transport is dominated by the barotropic component. The importance of this barotropic transport was also observed by Bindoff et al. (2000) and Heywood et al. (1998) to the east and west of the BROKE-West survey region where the ASC carries 29.4 ± 14.7 and 14 ± 3 Sv of total transport at 80−150°E and 15°W, respectively, while only observing baroclinic transports of 4.1 ± 3.9 and 3 Sv. Fahrbach et al. (1994) similarly observes bottom referenced baroclinic transports in the Weddell Gyre associated with the ASF of 3–11 Sv, but notes that only using bottom referenced baroclinic transport may miss 50–100% of the SAC transport (Fahrbach et al., 1991). Park et al. (2001) also observe that there is significant westward barotropic transport south of 65°S at 30°E. Our findings support these studies and extend their result between 30 and 80°E.

Finally, we note that the LADCP-derived transports across the sections produce mass balances that are unrealistically large, where we may have expected mass conservation around the hydrographic sections to within the observed error. The random instrument errors are not large enough to explain the imbalances of up to 75 Sv alone. Experiments combining LADCP referenced thermal wind transports lead to even larger mass imbalances (not shown in this paper). It seems likely that LADCP data have intrinsically different horizontal and vertical scales from the thermal wind estimates of the baroclinic transports, and that mixing these two data types (LADCP and CTD estimates) can lead to more strongly inconsistent transport estimates. While it is likely that temporal aliasing is a major contributor to the non-conservation of mass, we also observe that the direction of the LADCP absolute transport may be significantly different from the baroclinic component (Fig. 9). This is particularly evident at Legs 3 and 9, where the LADCP derived transport is in the opposite direction to the baroclinic transport north of the ASF, and is also apparent on the other legs where the eastward absolute transport north of the ASF is greater than the baroclinic component by over 8 Sv. These differences probably result from the different horizontal and vertical spatial scales of sampling used by the two methods. The geostrophic component of the LADCP absolute transport is the *in situ* velocity related to the local density gradient at each station, while the baroclinic transport is an estimate of the horizontal gradient between two stations separated by up to

75 km. The length scale difference results in a smoother, more filtered, baroclinic transport and a spatially aliased LADCP transport. The large distances that are integrated over north of the ASF to calculate transports exacerbates the spatial bias in the LADCP estimates of transport compared with the baroclinic estimate, and is the likely cause to the large divergences observed in this study. That the long length-scale baroclinic component is not reflective of the local absolute transports suggests that traditional CTD surveys may present a view of the ocean that is substantially over-smoothed and misses much of the oceanographic fine structure that is captured by the LADCP but is, on the other hand, a more reliable estimate of large scale integrated quantities such as transport and fluxes of heat and freshwater.

8. Conclusions

While there are many similarities between this survey and the earlier adjoining BROKE survey between 80 and 150°E, there are also some significant differences. In the BROKE survey there was only one "sub-polar gyre" located over the Australian-Antarctic basin and its location and size is largely defined by bathymetry and the ACC. In this survey there at least two gyres or recirculations (the eastern part of the Weddell Gyre and the Prydz Bay gyre). Relative to the BROKE survey the horizontal circulation is more complex, and the general influence of the bathymetry on circulation is less obvious. However the influence of the ACC in the eastern part of this survey is more distinct in defining the northern and eastern limit of these "subpolar gyres" (Fig. 7).

Similar to the earlier BROKE survey is the presence of local bottom water formation, in this case most likely originating west of Prydz Bay and probably associated with the deep depression beneath the Cape Darnley polynya. It is very likely that this source region is strongly seasonal, as observed in the Adélie Depression where Adélie Land Bottom Water is formed, and in that case represents 0.5–2.0 Sv (or upto 25%) of the bottom formation around Antarctica (Williams et al., 2008a).

While it is unlikely that this source at between 60 and 70°E is as large as the Adélie Land source, it does raise the important question of how to more accurately account for the contributions of these small and seasonally varying bottom water sources in the overall formation of bottom water around Antarctica.

Acknowledgments

We thank the officers and crew of the RSV *Aurora Australis* and support staff for their professionalism and hard work during the course of this 71 day voyage. Mark Rosenberg is thanked for his tireless effort in processing the CTD, ship ADCP and LADCP data. This work is supported by the Australian Governments Cooperative Research Centers Programme, through the Antarctic Climate and Ecosystems CRC.

Appendix A

A.1. Calculation of sea surface height

To visualize and understand the circulation regimes in the survey region a dynamic topography was calculated using CTD and LADCP data. The bottom referenced baroclinic velocity data were referenced to the bottom LADCP velocities which are assessed as the most reliable estimate of absolute velocity available, to create absolute surface velocities.

From these surface velocities we derived the surface height changes between stations using the relation

$$dh = \frac{f}{g} * \boldsymbol{U} * r \tag{A.1}$$

where $\boldsymbol{U}$ is the normal surface velocity estimate and r is the distance between the stations. The surface height changes were then integrated around each of the five closed ship track paths against the coast from west to east starting with zero at the coast. Due to measurement errors the surface height does not return to zero height at the eastern end of each loop. This non-closure of the path integral implies a flow through the coast. To correct for this we used a least-squares inversion with the constraint that there is no flow into the coast to redistribute the error in surface height over each cast in a self-consistent way. This corrected surface height field is shown in Fig. 7 and ensures both no flow across the Antarctic coastline and to first order the conservation of mass.

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